Experiment: 6 Date:27-7-26

5G NR Frame Structure Analysis with Different Subcarrier Spacing

Aim:

To analyze the 5G NR frame structure by studying the effect of different subcarrier spacings, slot durations, and OFDM symbols using MATLAB.

Objectives

1. To analyze different Subcarrier Spacing (SCS) values used in 5G NR numerology.
2. To compare the slot duration corresponding to different Subcarrier Spacing values in 5G NR.
3. To study the number of OFDM symbols present in a 5G NR slot for normal cyclic prefix.

Software Required

• MATLAB

Theory:

Unlike previous generations of mobile communication, 5G New Radio (NR) employs flexible numerology, where different subcarrier spacings are used depending on the deployment scenario
and service requirements. Increasing the subcarrier spacing reduces the slot duration, enabling lower latency and supporting diverse applications such as enhanced Mobile Broadband (eMBB),
Ultra-Reliable Low-Latency Communications (URLLC), and massive Machine-Type Communications (mMTC). Regardless of the selected numerology, a normal cyclic prefix slot contains 14 OFDM symbols.

Objective – 1

Analyze the standard subcarrier spacing (SCS) values defined in the 5G NR numerology using
MATLAB. Compare the relationship between the numerology index (μ) and subcarrier spacing to
understand the flexible bandwidth allocation employed in 5G communication systems.

Procedure:

• Define the supported 5G NR subcarrier spacing values.
• Display the values using MATLAB.
• Plot the subcarrier spacing for different numerologies.

Objective – 1: Analyze Different Subcarrier Spacing Values

MATLAB Program

```
clc;
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
mu = 0:4; % Numerology Index
bar(scs)
grid on
set(gca,'XTick',1:5)
set(gca,'XTickLabel',mu)
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing for Different Numerologies')
```

Expected Output

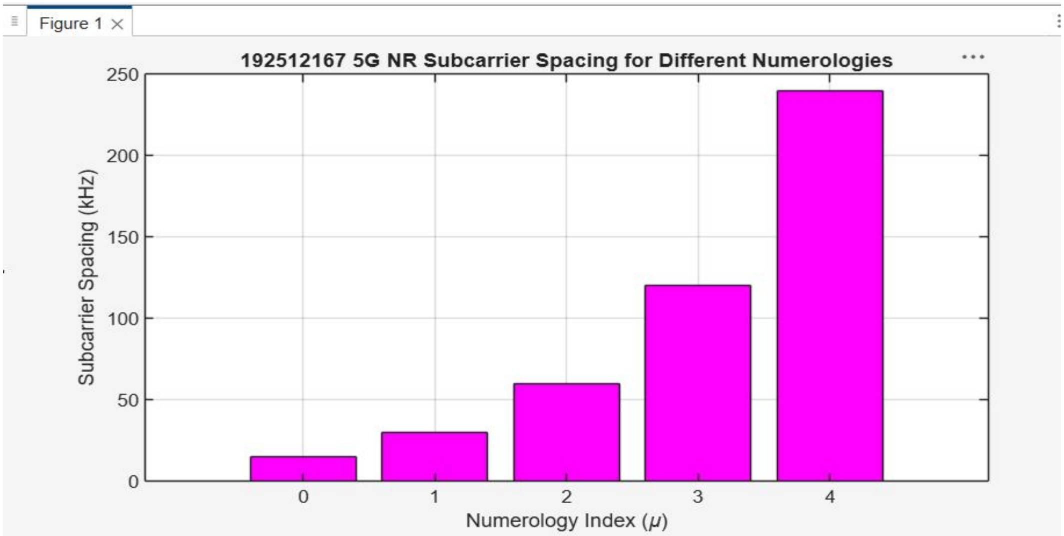
• Bar graph showing:
o 15 kHz
o 30 kHz
o 60 kHz
o 120 kHzo 240 kHz

exp21.m × exp22.m × exp53.m × exp6.m × +

/MATLAB Drive/exp6.m

```
clc;
clear;
close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
mu = 0:4; % Numerology Index
bar(scs,"magenta")
grid on
set(gca,'XTick',1:5)
set(gca,'XTickLabel',mu)
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('192512167 5G NR Subcarrier Spacing for Different Numerologies')
```



Objective – 2 Compare Slot Duration

Analyze the variation in slot duration corresponding to different 5G NR subcarrier spacing values
using MATLAB. Compare the inverse relationship between subcarrier spacing and slot duration
to understand how flexible numerology enables low-latency communication in 5G networks.

Procedure

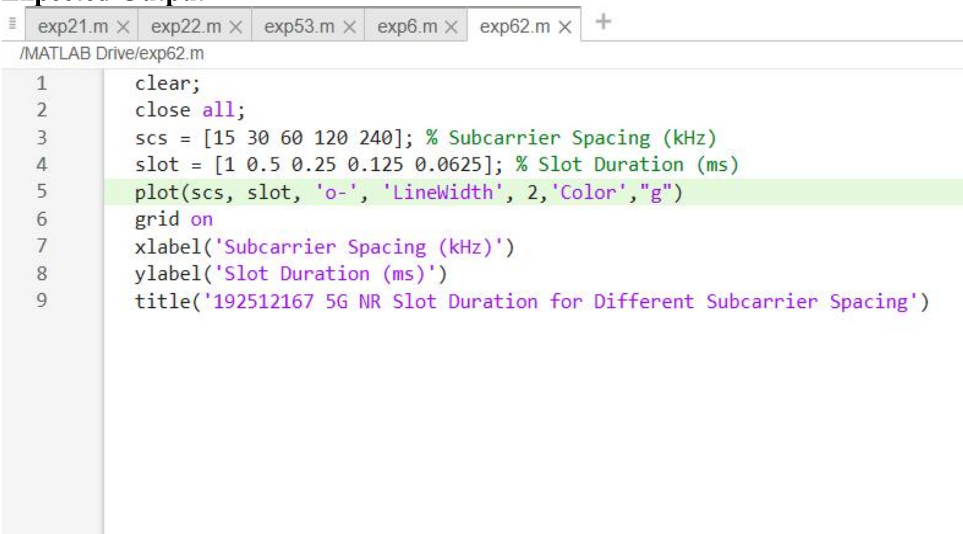
- Define the slot duration corresponding to each subcarrier spacing.
- Plot slot duration versus subcarrier spacing.
- Observe the inverse relationship.

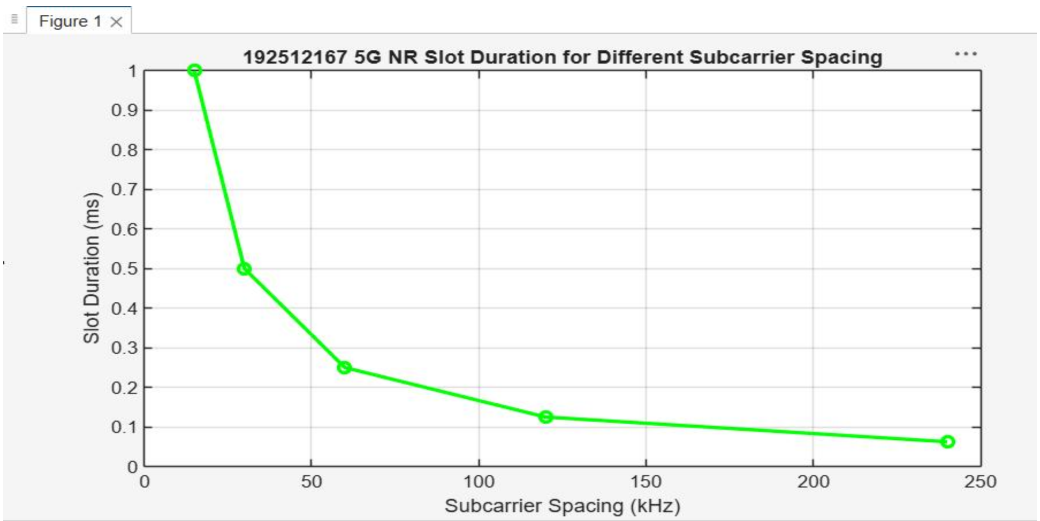
MATLAB Program

```
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
slot = [1 0.5 0.25 0.125 0.0625]; % Slot Duration (ms)
plot(scs, slot, 'o-', 'LineWidth', 2)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Slot Duration (ms)')
```

title('5G NR Slot Duration for Different Subcarrier Spacing')

Expected Output





• Graph showing slot duration decreasing as subcarrier spacing increases.

Objective – 3 Study the Number of OFDM Symbols

Analyze the number of OFDM symbols present in a 5G NR slot for different numerology configurations using MATLAB. Observe that, under the normal cyclic prefix, each slot consistently contains 14 OFDM symbols irrespective of the subcarrier spacing.

Procedure

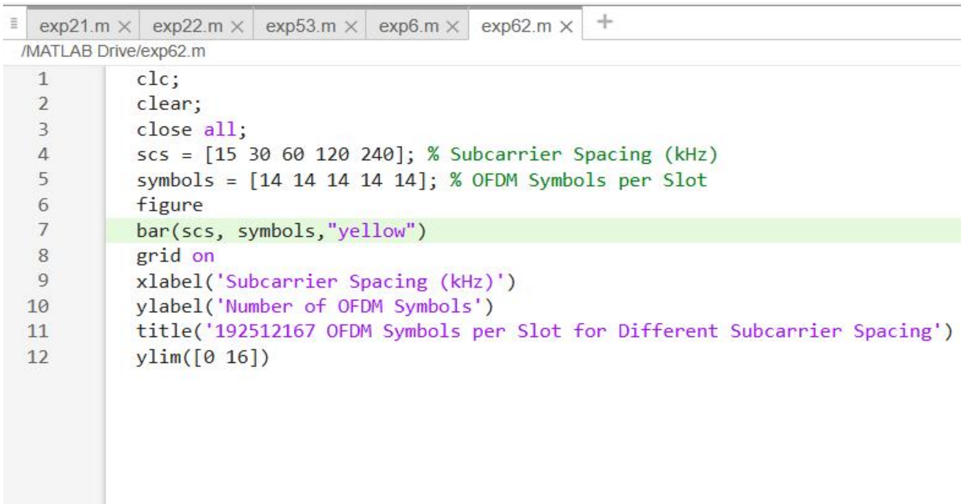
• Define the number of OFDM symbols in a slot.
• Plot the OFDM symbols for different numerologies.
• Observe that the number of symbols remains constant.

MATLAB Program

```
clc;
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
symbols = [14 14 14 14 14]; % OFDM Symbols per Slot
figure
bar(scs, symbols)
grid on
```

xlabel('Subcarrier Spacing (kHz)')
ylabel('Number of OFDM Symbols')
title('OFDM Symbols per Slot for Different Subcarrier Spacing')
ylim([0 16])

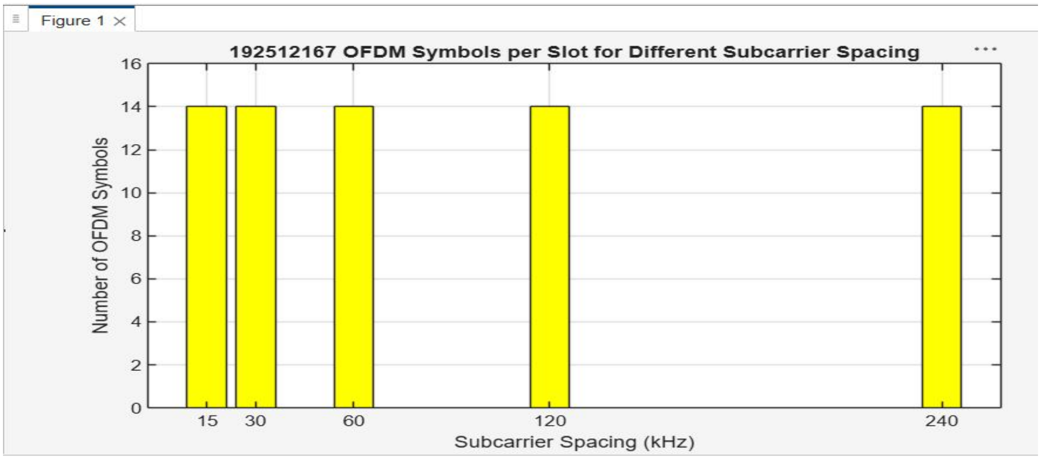
Expected Output



exp21.m exp22.m exp53.m exp6.m exp62.m

/MATLAB Drive/exp62.m

```
clc;
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
symbols = [14 14 14 14 14]; % OFDM Symbols per Slot
figure
bar(scs, symbols,"yellow")
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Number of OFDM Symbols')
title('192512167 OFDM Symbols per Slot for Different Subcarrier Spacing')
ylim([0 16])
```



Result

The MATLAB analysis successfully demonstrated the flexible numerology of 5G NR. The results
showed that increasing the subcarrier spacing decreases the slot duration while maintaining